Linear Regression and OLS

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

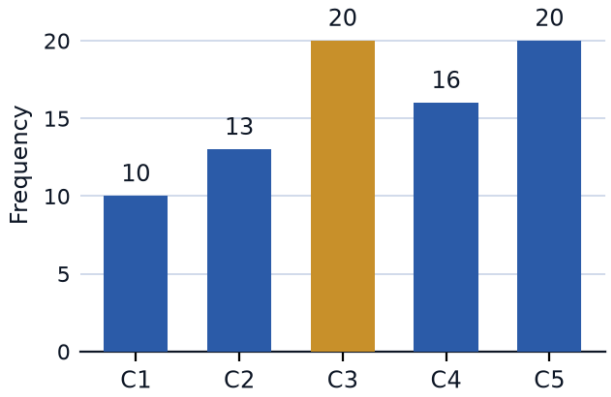
1 **[Foundational]** • **Target: Linear Regression & OLS Practice Exercises**

Solve the initial value problem for the coupled system of linear differential equations:

$\mathbf{x}'(t) = (1, 4; 2, 3)\mathbf{x}(t), \qquad \mathbf{x}(0) = (5; 1).$

2 **[Foundational]** • **Target: Linear Regression & OLS Practice Exercises**

Level 3 · Advanced / University RigorFind the general solution in parametric vector form for the underdetermined linear system: $x_1 + 2x_2 - 3x_3 + x_4 + 2x_5, = 3; -x_1 - 2x_2 + 4x_3 - 2x_4 - x_5, = -2; 2x_1 + 4x_2 - 5x_3 + x_4 + 5x_5, = 7$



3 **[Foundational]** • **Target: Linear Regression & OLS Practice Exercises**

Level 3 · AdvancedA scatterplot displays a point at $x = 18$, where the rest of the sample data has x-values between 2 and 8. The point falls almost exactly along the linear trend established by the other points. Explain the effect of this point on: (a) leverage, (b) the residual, (c) the correlation coefficient r, and (d) the slope b_1 of the LSRL.

4 **[Foundational]** • **Target: Linear Regression & OLS Practice Exercises**

Find the standard matrix $[T]$ of the linear operator $T : \mathbb{R}^2 \to \mathbb{R}^2$ that performs a horizontal shear by factor 2, followed by a counterclockwise rotation of 45°.

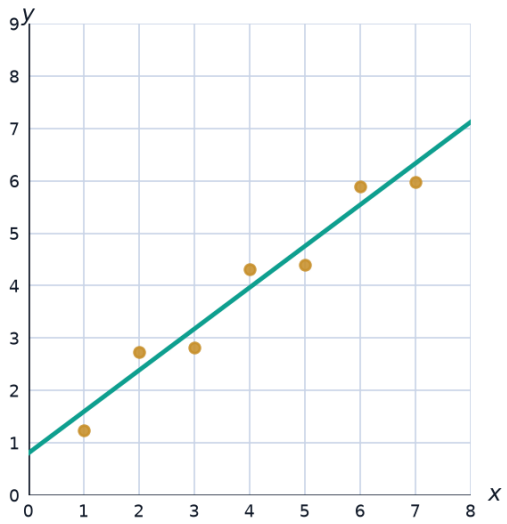
5 **[Foundational]** • **Target: Linear Regression & OLS Practice Exercises**

Level 3 · Advanced / University RigorLet $A = (2, 1; 0, 3)$. Using the Cayley–Hamilton Theorem, express A^4 and A^{-1} as linear combinations $c_1 A + c_0 I$.

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6 [Foundational] • Target: Linear Regression & OLS Practice Exercises

Using the regression model $\hat{y} = 38.0 + 0.55x$, a student who completed 80% of homework scored 86 on the exam. (a) Calculate the residual. (b) Explain what a random scatter of points around 0 on a residual plot indicates about model adequacy.



7 [Foundational] • Target: Linear Regression & OLS Practice Exercises

Level 2 · MediumFind all solutions to the homogeneous linear system:

$x_1 + 3x_2 - 2x_3, = 0; 2x_1 + 6x_2 - 4x_3, = 0$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

A least-squares regression line predicting exam score (y) from homework completion percentage (x) is $\hat{y} = 38.0 + 0.55x$ with correlation $r = 0.82$. (a) Interpret the slope 0.55 in context. (b) Calculate and interpret the coefficient of determination r^2.

9 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

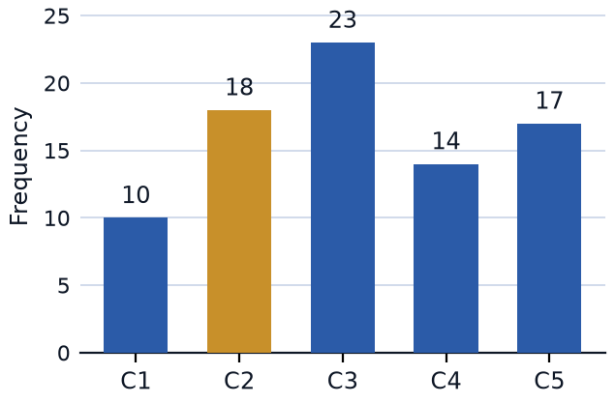
Level 1 · FoundationsConvert the matrix into Reduced Row Echelon Form (RREF):

$M = (1, -2, 1; 2, -4, 5).$

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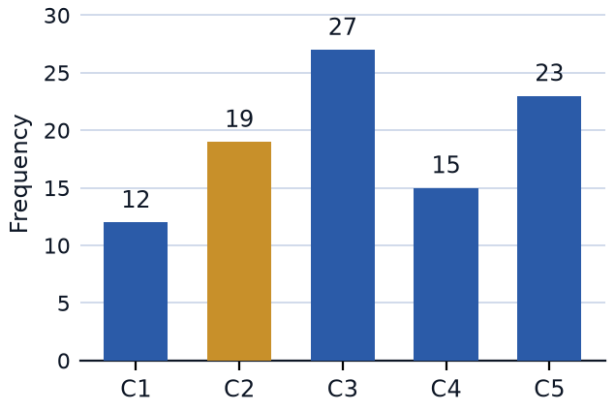
10 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

A linear transformation $T:\mathbb{R}^7\to\mathbb{R}^4$ has $\dim(\ker T)=3$. What is the dimension of the range of T, and is T surjective?



11 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

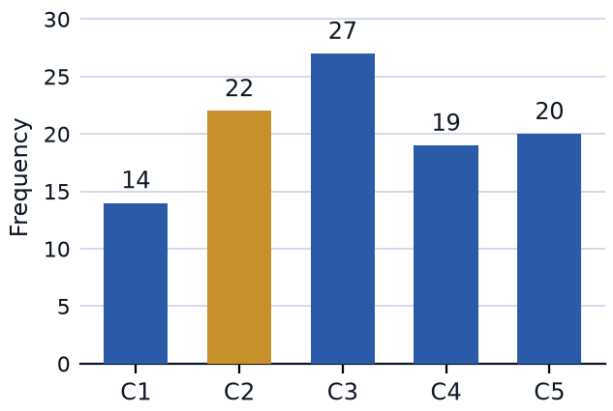
Let $f(x,y)=x^3e^{2y}-y\ln(x)$ defined on $x>0$. Compute the first-order partial derivatives $f_x(x,y)$ and $f_y(x,y)$, and evaluate $f_x(1,0)$ and $f_y(1,0)$.



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12 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

Classify the critical points $(0, 0)$ and $(1, 1)$ of $f(x, y) = x^3 + y^3 - 3xy$ using the Second Derivative Test and Hessian matrix.



13 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

Use Stokes' Theorem to evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F} = \langle -y^3, x^3, -z^3 \rangle$ and C is the intersection of the cylinder $x^2 + y^2 = 1$ and the plane $z = 1$, oriented counterclockwise when viewed from above. Verify surface independence by comparing the flux of $abla \times \mathbf{F}$ through the flat disk $S_1 : x^2 + y^2 \le 1, z = 1$ and the paraboloid cap $S_2 : z = 2 - (x^2 + y^2)$.

14 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

Find all eigenvalues, their algebraic and geometric multiplicities, and a basis for each eigenspace for: $A = (3, 1, 0; 0, 3, 0; 0, 0, 2)$. Is A diagonalizable?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

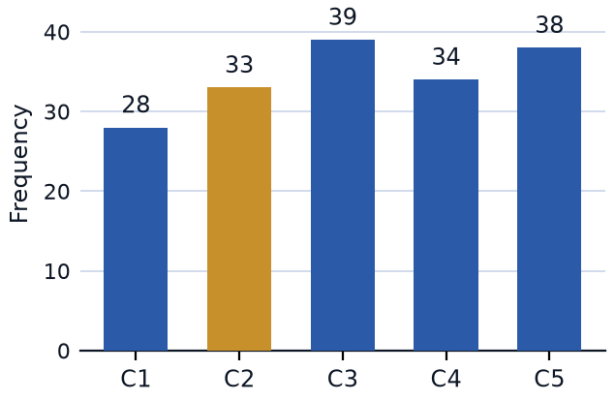
15 [Challenging] • Target: Linear Regression & OLS Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification

Apply the Gram–Schmidt orthogonalization process to the columns of A to find its QR factorization: $A = (1, 2; 2, 1; -2, 2)$.

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16 **[Challenging]** • **Target: Linear Regression & OLS Practice Exercises** • **Cross-Lesson: 8.01 PCA and Dimensionality Reduction**

Use the Divergence Theorem to calculate the outward flux of $\mathbf{F} = \langle x^3 + y\sin z,\ y^3 + z\cos x,\ z^3 + xy \rangle$ across the closed boundary surface of the solid cylinder $E : x^2 + y^2 \le a^2,\ 0 \le z \le h$.



17 **[Challenging]** • **Target: Linear Regression & OLS Practice Exercises** • **Cross-Lesson: 1.01 Vectors and Embeddings in AI**

Prove that if λ is an eigenvalue of an invertible matrix A, then $\lambda eq 0$ and λ^{-1} is an eigenvalue of A^{-1} with the exact same corresponding eigenvector.

18 **[Challenging]** • **Target: Linear Regression & OLS Practice Exercises** • **Cross-Lesson: 2.01 Neural Layer Matrices and Activations**

Prove that the 3D vector field $\mathbf{F} = \langle 2xy + z^2,\ x^2 + 2yz,\ 2xz + y^2 \rangle$ is conservative on $\mathbb{R}^3$, construct a potential function f satisfying $ablaf = \mathbf{F}$, and evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(0, 0, 0)$ to $(1, 2, 3)$.

19 **[Challenging]** • **Target: Linear Regression & OLS Practice Exercises** • **Cross-Lesson: 3.01 Gradient Descent and Optimization**

Diagonalize the matrix $A = (1, 2; 2, 1)$ by finding invertible P and diagonal D such that $A = PDP^{-1}$. Then compute A^k for any positive integer k.

20 **[Challenging]** • **Target: Linear Regression & OLS Practice Exercises** • **Cross-Lesson: 4.01 Conditional Probability and Bayes Rule**

Level 3 · AdvancedFind the coordinates of all points on the curve $x^2 - xy + y^2 = 9$ where the tangent line is horizontal.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Linear Regression & OLS Practice Exercises

A biostatistician is investigating the relationship between the dosage of a new experimental drug (measured in milligrams) and the reduction in systolic blood pressure (in mmHg) for patients with mild hypertension. They collected data from $n = 200$ patients and fit a simple linear regression model. The model's output yielded an estimated intercept of $\hat{\beta}_0 = 5.2$ and an estimated slope of $\hat{\beta}_1 = -0.8$. The standard error of the estimated slope is $\text{SE}(\hat{\beta}_1) = 0.15$. A 95% confidence interval for the true slope β_1 was calculated to be $[-1.12, -0.48]$. The biostatistician then constructs a hypothesis test to determine if the true slope is significantly different from zero. To do this, they calculate a test statistic. If the same α level used for the confidence interval were applied to a two-tailed hypothesis test of $H_0 : \beta_1 = 0$ versus $H_a : \beta_1 eq 0$, what would be the approximate p-value associated with this test statistic?

a. Greater than 0.05

b. Equal to 0.05

c. Less than 0.05

d. Cannot be determined from the given information

22 [Foundational] • Target: Linear Regression & OLS Practice Exercises

Consider a linear transformation represented by the matrix $A = (3, 0; 0, -2)$. If $\mathbf{v} = (1; 1)$ is an eigenvector of A, what is the corresponding eigenvalue?

a. 3

b. -2

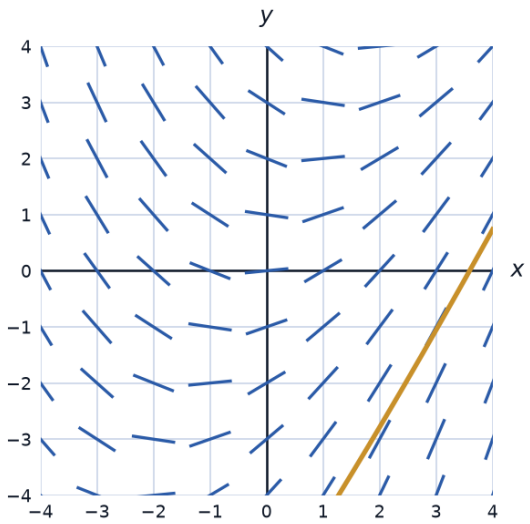
c. 1

d. 5

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23 [Foundational] • Target: Linear Regression & OLS Practice Exercises

Consider a linear transformation represented by the matrix $A = (2, 1; 1, -2)$. A point P is mapped to P' by this transformation. If the coordinates of P are $(x; y)$ and the coordinates of P' are $(x'; y')$, then $(x'; y') = A(x; y)$. Suppose a curve is defined by the equation $x'^2 + y'^2 = 5$. What is the equation of the original curve $x^2 + y^2 = k$ in terms of x and y that transforms into this circle? Find the value of k.



a. 1

b. 2

c. 3

d. 4

24 [Foundational] • Target: Linear Regression & OLS Practice Exercises

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation represented by the matrix $A = (3, -1\ 1, 2)$ with respect to the standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$. Consider a new basis $B = \{\mathbf{v}_1, \mathbf{v}_2\}$ where $\mathbf{v}_1 = (1\ 1)$ and $\mathbf{v}_2 = (1\ -1)$. Let P be the change-of-basis matrix from basis B to the standard basis. Find the matrix representation of T with respect to basis B, denoted by $[T]_B$. This matrix can be expressed as $P^{-1}AP$. Find the sum of the diagonal entries of $[T]_B$.

a. 5

b. 6

c. 4

d. 7

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25 [Foundational] • Target: Linear Regression & OLS Practice Exercises

Consider a linear transformation $T: \mathbb{R}^3 \to \mathbb{R}^2$ defined by $T(x, y, z) = (2x - y + z, x + 3y - 2z)$. Let V be the image of T, i.e., $V = \text{Im}(T)$. Furthermore, let W be the subspace of $\mathbb{R}^2$ defined by $W = \{(a, b) \in \mathbb{R}^2 \mid a + b = 0\}$. Find the dimension of the intersection of V and W, i.e., $\dim(V \cap W)$.

a. 0

b. 1

c. 2

d. Cannot be determined

26 [Foundational] • Target: Linear Regression & OLS Practice Exercises

Let $T: \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation such that $T(x, y) = (3x - y, x + 2y)$. Consider the standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$ for $\mathbb{R}^2$, where $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$. Let $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$ be another basis for $\mathbb{R}^2$, where $\mathbf{v}_1 = (1, 1)$ and $\mathbf{v}_2 = (1, -1)$. If the matrix representation of T with respect to the basis $\mathcal{B}$ is $[T]_{\mathcal{B}} = (a, b\ c, d)$, and $a + d = 5$, determine the value of $\det(T)$.

a. 7

b. 5

c. 3

d. 2

27 [Foundational] • Target: Linear Regression & OLS Practice Exercises

A social scientist is investigating the relationship between the number of hours spent studying per week (X) and the final exam score (Y) for a group of $n = 100$ students. The least-squares regression line is given by $\hat{y} = 55.2 + 8.5x$. The standard error of the slope is $SE_b = 1.2$. The researcher wants to construct a 95% confidence interval for the true slope of the regression line. What is the value of the t-score used for this confidence interval?

a. 1.645

b. 1.960

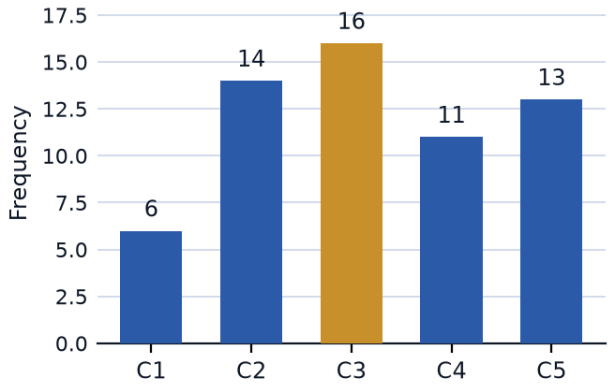
c. 1.984

d. 2.576

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28 [Foundational] • Target: Linear Regression & OLS Practice Exercises

A solid object in $\mathbb{R}^3$ is defined by the region bounded by the planes $x = 0$, $y = 0$, $z = 0$, and the plane passing through the points $P = (6, 0, 0)$, $Q = (0, 4, 0)$, and $R = (0, 0, 3)$. If a linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ is represented by the matrix $M = (1, 0, 0; 0, 1, 0; 0, 0, 2)$, what is the volume of the transformed solid object?



a. 12

b. 24

c. 36

d. 48

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation represented by the matrix $A = (2, k\ 1, 3)$ with respect to the standard basis. The image of the vector $\mathbf{v} = (1\ 1)$ under T is $T(\mathbf{v}) = (3\ 4)$. Find the value of the parameter k.

a. 1

b. 2

c. 3

d. 4

30 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

Consider the linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ defined by $T(\mathbf{x}) = A\mathbf{x}$, where $A = (1, 2, 3; 4, 5, 6; 7, 8, 9)$. Let V be the image of T, i.e., $V = \text{Im}(T)$. V is a subspace of $\mathbb{R}^3$. Find the dimension of the orthogonal complement of V, denoted by $V^\perp$.

a. 1

b. 2

c. 0

d. 3

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31 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

Consider the following system of linear equations represented by the augmented matrix: $(1, 2, |, 5; 2, -1, |, 0)$. Use row reduction to transform the matrix into row-echelon form and determine the unique solution (x, y) for the system. What is the value of x?

a. 1

b. 2

c. 3

d. 4

32 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

A linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ is defined by $T(x, y) = (ax + by, cx + dy)$. The matrix representation of T with respect to the standard basis is $A = (a, b\ c, d)$. The image of the standard basis vectors $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$ under T are given by $T(\mathbf{e}_1) = (3, -1)$ and $T(\mathbf{e}_2) = (1, 2)$. The transformation is applied to a square with vertices $(0, 0), (1, 0), (0, 1), (1, 1)$. The area of the transformed square is known to be k. Find the value of k.

a. 5

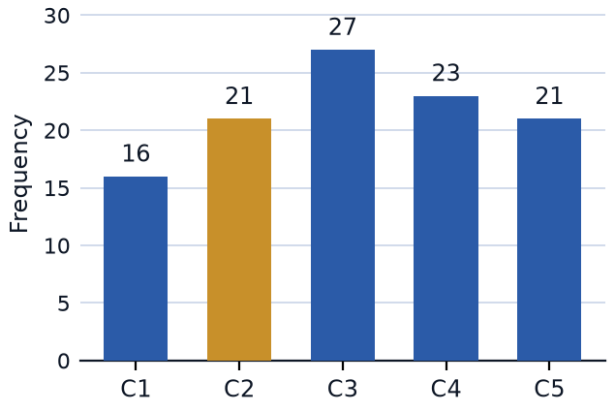
b. 10

c. 15

d. 20

33 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

Consider the linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ represented by the matrix $A = (3, 1; 1, 3)$. Identify one of the eigenvalues of A.



a. 1

b. 2

c. 3

d. 4

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34 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

Consider the linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ represented by the matrix $A = (1, 2; k, 4)$. The image of the vector $\mathbf{v} = (1; 1)$ under this transformation is $T(\mathbf{v}) = (3; 5)$. Determine the value of the parameter k.

a. 1

b. 2

c. 3

d. 4

35 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

Consider the linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ represented by the matrix $A = (2, 0\ 0, 3)$. What is the image of the vector $\mathbf{v} = (1\ 1)$ under this transformation, i.e., find $T(\mathbf{v})$?

a. $(1\ 1)$

b. $(2\ 3)$

c. $(3\ 2)$

d. $(4\ 9)$

36 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

Consider the linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ represented by the matrix $A = (1, 2; 3, 4)$. If a vector $\mathbf{v} = (2; 1)$ is transformed by T, what is the resulting vector $T(\mathbf{v})$?

a. $(4; 10)$

b. $(2; 3)$

c. $(4; 7)$

d. $(3; 5)$

37 [Intermediate] • Target: Linear Regression & OLS Practice Exercises

The scatterplot below displays the relationship between the number of hours (h) students spent studying for a math test and their scores (s) on that test. The equation of the least-squares regression line for this data is given by $s = 1.2h + 78.5$. If a student studied for 5 hours, what is their predicted score based on this model?

a. 91.5

b. 84.5

c. 81.5

d. 90.5

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Linear Regression & OLS Practice Exercises • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

A linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ is defined by $T(\mathbf{x}) = A\mathbf{x}$, where $A = (2, 1; 1, 3)$. If $\mathbf{v} = (1; 2)$, find $T(\mathbf{v})$.

a. $(4; 7)$

b. $(3; 7)$

c. $(4; 5)$

d. $(5; 7)$

39 [Challenging] • Target: Linear Regression & OLS Practice Exercises • Cross-Lesson: 3.01 Gradient Descent and Optimization

Consider the following system of linear equations represented by an augmented matrix: $[1, 2, |, 5; 0, 1, |, 2]$. What is the value of x in the first equation?

a. 1

b. 2

c. 3

d. 4

40 [Challenging] • Target: Linear Regression & OLS Practice Exercises • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

Consider the system of linear equations represented by the following augmented matrix: $(1, 2, |, 5; 0, 1, |, 2)$ Use row reduction to determine the unique solution for x and y in the system. What is the value of x?

a. 1

b. 2

c. 3

d. 4

41 [Challenging] • Target: Linear Regression & OLS Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

The scatterplot below displays the relationship between the number of hours students spent studying for a test and their scores on that test. The equation of the least-squares regression line is approximately $\hat{y} = 55.2 + 4.8x$, where x is the number of hours spent studying and $\hat{y}$ is the predicted test score.

a. 55.2

b. 4.8

c. 50

d. 70

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42 [Challenging] • Target: Linear Regression & OLS Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification

Which of the following is a subspace?

a. x

b. $\{p \in P_2 : p(0) = 1\}$

c. x

d. $\{A \in M_{2\times 2} : \det A = 0\}$

43 [Challenging] • Target: Linear Regression & OLS Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

A linear map T satisfies $T\langle -1, -1\rangle = \langle 0, -2\rangle$ and $T\langle 0, -1\rangle = \langle 2, -1\rangle$. Find $T\langle 2, -3\rangle$.

a. $\langle -6, -1\rangle$

b. $\langle -10, 1\rangle$

c. $\langle 10, -1\rangle$

d. $\langle 11, -1\rangle$

44 [Challenging] • Target: Linear Regression & OLS Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI

Find the image of $\langle 1, 1\rangle$ under the rotation counterclockwise by 30°.

a. $\langle -\frac{1}{2} + \frac{\sqrt{3}}{2}, \frac{1}{2} + \frac{\sqrt{3}}{2}\rangle$

b. $\langle \frac{1}{2} + \frac{\sqrt{3}}{2}, -\frac{1}{2} + \frac{\sqrt{3}}{2}\rangle$

c. $\langle \frac{1}{2} - \frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2} - \frac{1}{2}\rangle$

d. $\langle 1, 1\rangle$

45 [Challenging] • Target: Linear Regression & OLS Practice Exercises • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

Solve $y' - \frac{y}{x} = x^2$ (linear first-order):

a. $y = \frac{1}{2}x^3 + Cx$

b. $y = \frac{1}{3}x^3 + C$

c. $y = x^3 + \frac{C}{x}$

d. $y = x^3 + Cx$

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